

Lesson 14: Systems of Differential Equations - Matrices & Stability

Professor Dave Explains Differential Equations Master Series · Lesson 14

Video: Systems of Differential Equations Part 2: Matrices and Stability Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that matrix algebra is the native language of multivariable calculus. By packaging states into a vector $\mathbf{x}(t)$ and interaction rates into a matrix $\mathbf{A}$, the system becomes $\mathbf{x}' = \mathbf{A}\mathbf{x}$. The eigenvalues λ represent the natural frequencies or growth rates of the system, while the eigenvectors $\mathbf{v}$ define the invariant directional axes of phase space. The trace and determinant of $\mathbf{A}$ instantly classify whether the origin is a stable sink, unstable source, or twisting spiral.

2. Core Theoretical Framework & Rigorous Formulations

1. Matrix Eigensystem Formulation

$$\mathbf{x}' = \mathbf{A}\mathbf{x}, \quad \det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 - \tau\lambda + \Delta = 0$$

where $\tau = \text{tr}(\mathbf{A}) = \lambda_1 + \lambda_2$ and $\Delta = \det(\mathbf{A}) = \lambda_1\lambda_2$. General solution: $\mathbf{x}(t) = c_1e^{\lambda_1t}\mathbf{v}_1 + c_2e^{\lambda_2t}\mathbf{v}_2$.

2. Phase Plane Critical Point Classification

Eigenvalue Spectrum	Critical Point Type	Stability Regime
$\lambda_1 < \lambda_2 < 0$ (Both Real Negative)	Nodal Sink	Asymptotically Stable
$\lambda_1 > \lambda_2 > 0$ (Both Real Positive)	Nodal Source	Unstable
$\lambda_1 < 0 < \lambda_2$ (Opposite Signs, $\Delta < 0$)	Saddle Point	Unstable (Hyperbolic)
$\alpha \pm i\beta$ with $\alpha < 0$	Spiral Sink	Asymptotically Stable
$\alpha \pm i\beta$ with $\alpha > 0$	Spiral Source	Unstable
$\pm i\beta$ (Pure Imaginary, $\tau = 0, \Delta > 0$)	Center (Elliptic orbits)	Neutrally Stable

3. Fully Solved Whiteboard Problems

Exam Problem: Complete Eigen-Decomposition and Phase Portrait Analysis

Problem Statement: Solve $\mathbf{x}' = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \mathbf{x}$, identify eigenvalues and eigenvectors, and classify the stability of the origin.

Step 1: Compute Characteristic Polynomial

$$\tau = 1 + 2 = 3, \Delta = (1)(2) - (2)(3) = 2 - 6 = -4. \det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 - 3\lambda - 4 = (\lambda - 4)(\lambda + 1) = 0.$$

Step 2: Find Eigenvalues

$$\lambda_1 = 4, \quad \lambda_2 = -1.$$

Step 3: Compute Eigenvectors

For $\lambda_1 = 4$: $(\mathbf{A} - 4\mathbf{I})\mathbf{v}_1 = \begin{pmatrix} -3 & 2 \\ 3 & -2 \end{pmatrix} \mathbf{v}_1 = 0 \implies -3v_x + 2v_y = 0 \implies \mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$. For $\lambda_2 = -1$: $(\mathbf{A} + \mathbf{I})\mathbf{v}_2 = \begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix} \mathbf{v}_2 = 0 \implies v_x + v_y = 0 \implies \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Step 4: Formulate General Vector Trajectory

$$\mathbf{x}(t) = c_1e^{4t} \begin{pmatrix} 2 \\ 3 \end{pmatrix} + c_2e^{-t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Step 5: Stability Classification

Since $\lambda_2 = -1 < 0 < \lambda_1 = 4$ (determinant $\Delta = -4 < 0$), the origin $(0, 0)$ is a **Saddle Point**. It is unstable; trajectories approach along the stable axis $\mathbf{v}_2$ and diverge along the unstable axis $\mathbf{v}_1$.

⚠ Common Exam Pitfall & Trapdoor Avoidance

If determinant $\Delta < 0$, the origin is AUTOMATICALLY an unstable saddle point! You do not even need to calculate the trace to confirm instability.

4. Key Takeaways & Exam Strategy Formula Card

🎯 Core Takeaway & Exam Rule

Eigenvalues dictate trajectory growth/decay rates; eigenvectors dictate the straight-line axes of phase space.

- Video reference: [Systems of Differential Equations Part 2: Matrices and Stability](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.